26900125 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSECND302 - FOOD COMMODITIES II

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. When evaluated using the water floatation test, what does it mean if an egg floats? [U] / [CO1]

   a). It is of good quality | b). It is poor quality | c). High moisture content of the yolk | d). None of these

2. Which of the following is a physical characteristic of fresh fish suitable for processing into products like FPC? [U] / [CO1]

   a). Convex, clear, and bulging eyes | b). Brownish gills with opaque mucus | c). Sunken eyes with opaque pupils | d). Wrinkled skin that shrinks from flesh

3. 10. Which two culinary preparations rely directly on the emulsifying properties of egg yolk? [K] / [CO1]

   a). Mayonnaise and hollandaise sauce | b). Sponge cakes and meringues | c). Quiches and puddings | d). Broths and consommés

4. The primary carbohydrates found in meat are: [U] / [CO2]

a). Glycogen and glucose | b). Starch and cellulose | c). Fructose and sucrose | d). Maltose and lactose

5. The ideal cooking method for meat should: [K] / [CO2]

a). Minimise the hardening of contractile protein | b). Maximise the hardening of contractile protein | c). Minimise connective tissue softening | d). None of these

6. ..... is a fermented sap of palm trees by natutal contamination. [U] / [CO3]

a). Rum | b). Toddy

c). Beer | d). Wine

7. ..... is the process of mixing sufficient carbon dioxide with water or beverage so that when served, the product gives off the gas in fine bubbles and the characteristic pungent taste or tang one feels. [U] / [CO3]

a). Fermentation | b). Carbonation

c). Pasteurization | d). Homogenization

8. The active principles of cinnamon are eugenol, cineole and ..... [K] / [CO3]

a). Trypsin | b). Cinnamaldehyde

c). Borneol | d). Piperine

9. Soyabean is classified under oilseeds mainly because of its high ..... [U] / [CO4]

a). Protein content | b). Fat content

c). Fibre content | d). Mineral content

10. Nuts and oilseeds are used in cookery in the form of ..... [U] / [CO4]

a). Whole, flaked and ground | b). Only whole form

c). Only powdered form | d). Only roasted form

[1x10 = 10]

Part B

Short Answer

Answer any **5** questions

Each question carries **2** marks

11. State two health benefits of consuming designer eggs. [K] / [CO1]

12. Explain the reason for supplementary value of fish with cereals and pulses. [U] / [CO1]

13. Describe Lysozyme. [U] / [CO1]

14. Explain why commercial ageing is sometimes performed at higher temperatures like 20°C. [U] / [CO2]

15. Describe what is fermentation of cocoa beans? Mention its importance. [U] / [CO3]

16. Explain the purpose of blending different teas. [U] / [CO3]

17. Discuss any four minerals present in nuts. [U] / [CO4]

[2x5 = 10]

Part C

Short Essay

Answer any **6** questions

Each question carries **5** marks

18. Describe any five factors affecting the coagulation of egg proteins. [U] / [CO1]

19. List out the different structures associated with the egg yolk and state their functions. [K] / [CO1]

20. Explain the distribution and physiological roles of B-complex vitamins and essential minerals found in poultry meat. [U] / [CO2]

21. Discuss the role of spices in cookery. [U] / [CO3]

22. Discuss on curry leaves. [U] / [CO3]

23. Discuss on the medicinal uses of ginger. [U] / [CO3]

24. Explain hydrogenation of vegetable oils and mention its effects on the properties of fat. [U] / [CO4]

25. Discuss the changes that occur in fats and oils on heating. [U] / [CO4]

[5x6 = 30]

Part D

Essay

Answer any **2** questions

Each question carries **10** marks

26. Discuss the mechanical and environmental factors that affect egg-white foam formation, focusing on beating time, beater type, container design, and temperature. [U] / [CO1]

27. Explain the mechanical dressing operations in poultry processing, focusing in detail on stunning, scalding, defeathering, evisceration, and chilling. [U] / [CO2]

28. Explain the processing of coffee and discuss any two products of coffee. [U] / [CO3]

29. Explain the characteristics and uses of groundnut, sesame, rice bran, rapeseed and mustard oils. [U] / [CO4]

[10x2 = 20]